EXECUTIVE SUMMARY

HOW FLIGHT CHANGES REACH ABU DHABI'S HOTELS

AUH Twin turns an airline schedule into hotel guest nights, by nationality and month. A planner can add a route, cut a market or change the transfer share, and see the guests it brings, with a range for good and bad years. It is built on DCT's own data and tested on months it never saw.

8.3%

error on daily hotel guests by nationality when check-ins are known, against 24.8% for last year's numbers

19.2%

error on guest nights by nationality and month **from flights alone**, against 24.4%; 7.6% on the monthly total

9.65M

guest nights from 45 nationalities over Aug 2025 to Jul 2026, traced back to the seats that bring them

THE TEAM

Team Odyssey

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Use of AI tools. AI tools helped us write code and draft text. The method, the tests and the results are ours.

THE PROBLEM

Flights and hotels count different people

Flight data knows where a plane took off.

A flight from London lands with 290 seats. Many on board are not British, and many fly straight on to another country.

Hotel data knows the guest's passport.

British guests arrive on London flights, but also through other hubs and by road. Nothing in the data links a seat to a guest.

So a planner cannot answer the central question: **if we add or lose this route, how many hotel nights change, for which nationality, and when?**

~50%

of passengers landing at AUH fly onward; the point-to-point share fell from 52.3% (2023) to 47.7% (early 2026)

12 of 45

nationalities in the hotel data have no direct flight to AUH at all

0.17 to 4.7

hotel guests per direct passenger, from India to China: one fixed ratio cannot work

OUR SOLUTION

One transparent chain from seats to guest nights

SEATS		PASSENGERS		VISITORS		HOTEL GUESTS		GUEST NIGHTS
14.64M	x 85.2% load factor	12.48M	x 48.9% point to point	6.10M	x 44.9% stay in hotels	2.74M	x 3.52 nights each	9.65M

Linking nationality to origin

For each nationality we fit, on 24 months of real data, how many guests come per point-to-point passenger on its own country's routes, how many come with the rest of the network, and how many arrive without an AUH flight:

```
guests = a_own x own P2P + a_other x other P2P + b
```

All three are kept non-negative, so the fit can never invent negative guests.

Anchored to what really happened

The chain predicts the **change**; last year's real guest nights give the **level**:

```
nights = last year's nights x chain(this month) /
chain(same month last year)
```

This one step cut the simulator's error from 27.7% to 19.2%. For daily figures with check-ins known, a median of four models (CatBoost, random forest, Prophet, Chronos-2) does the nowcast.

SYSTEM DESIGN

From the DCT data pack to a planner's screen

1. DATA

DCT hotel data

Daily guests and check-ins
45 nationalities + domestic
guests to Jul 2025,
check-ins to Feb 2026

DCT flight data

Seats, passengers, P2P, transfers
130 cities, 33 countries
Monthly 2022, daily 2023 to 2026

Open reference

OurAirports: city to country
and coordinates

2. MODEL ENGINE (PYTHON)

Nowcast

Median of 4 models
daily guests, 8.3% error

Simulator chain

seats > passengers > visitors
> hotel guests > guest nights
fitted per nationality,
anchored to last year

Back-test harness

pre-registered rules,
block bootstrap, WMAPE

3. SERVICES

FastAPI service

Planner

routes, seat budget, stress test
200 futures from real errors

Ask

Laya routes the question;
Gemini phrases, a guard checks

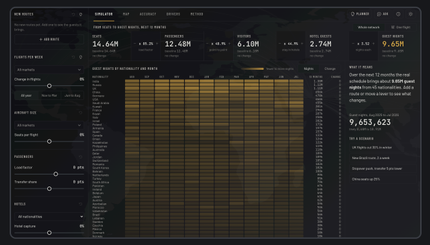
Scenario and accuracy

any lever change in
well under a second

4. PLANNER'S SCREEN

React dashboard

Simulator: chain and lattice
One flight seat map
Map of the real network
Planner dialog
Ask pane
Accuracy and drivers



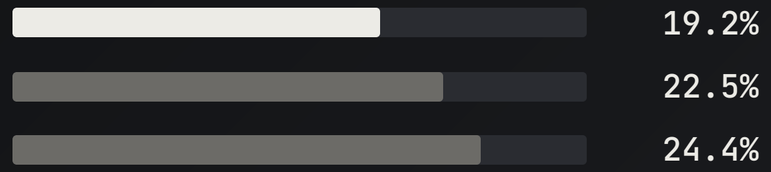
The Simulator view, live

Every number on screen
comes from the engine

ACCURACY: TESTED ON MONTHS THE MODEL NEVER SAW

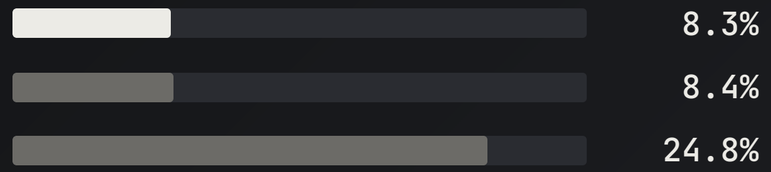
Better than last year's numbers, on every test we ran

SIMULATOR, FROM THE FLIGHT SCHEDULE ALONE



Guest nights by nationality and month, 7 months ahead, three test windows in 2024 and 2025. On the monthly total the error is **7.6%**.

NOWCAST, WITH THE DAY'S CHECK-INS KNOWN



Daily in-house guests for 45 nationalities and domestic; 16 approaches compared. Test-period predictions for Aug 2025 to Feb 2026 are ready.

WHY YOU CAN TRUST THE NUMBERS

Rules first. Every test was written down and committed before it ran, with a fixed rule for adopting a model.

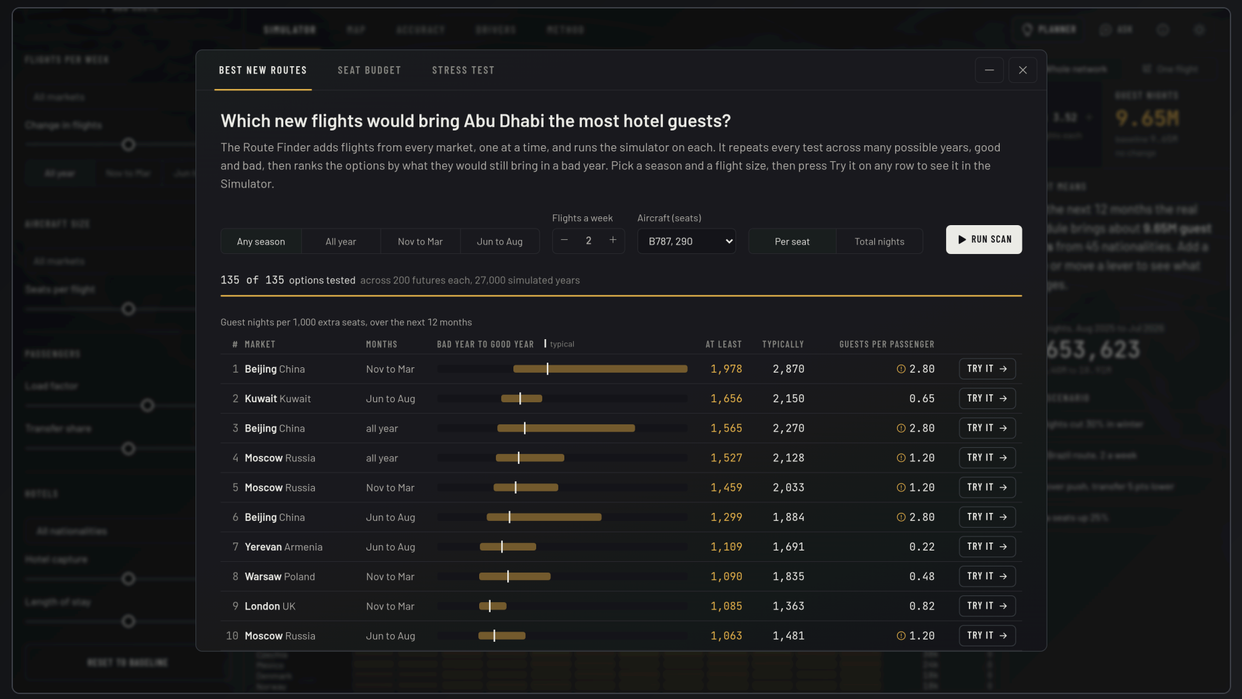
Failures reported. Our round-3 idea (shock-aware anchors) lost by 1.1 to 1.3 points. We kept the simpler model and published the result.

Outside check. The same method beats the standard baseline on 17 years of Singapore airport and visitor data: 5.6% against 8.8%.

Real data only. No synthetic or mocked inputs anywhere in the product. Error is WMAPE: the total miss divided by the total actual.

IMPACT

From what-if to what-to-do



The Planner tries every market and season across 200 possible years each (27,000 simulated years), and ranks by what a bad year still brings.

Which new route?

Ranked by guest nights per 1,000 extra seats in a bad year. Per seat, the best existing market brings `2,471` nights and the weakest `46`.

Where should extra seats go?

For `5,000` extra seats a week, the Planner builds a mix worth about `+380,000` guest nights a year, and at least `+283,000` in a bad year.

Which markets can we not afford to lose?

Losing all India flights would cost about `2.3M` guest nights a year, close to a quarter of the total. The UK and Russia follow.

Which lever matters most?

Transfer share. Turning 5 points of connecting passengers into visitors moves guest nights by about `11%`, compared to a 10% seat increase (`8.6%`).

FEASIBILITY

Runs on data DCT already has, on one laptop

WHAT IT TAKES

Data: the two feeds in this pack, hotel guests by nationality and flights into AUH by route. A monthly refresh re-fits the chain in under a minute.

Stack: Python model engine, FastAPI, a React dashboard. All open source; one command starts it, and it packages into a single container.

Speed: a full scenario answers in well under a second; a Planner scan of 135 options finishes in about 2 seconds.

Ask in plain English: the open Laya decision model routes the question to levers. Gemini only phrases the answer, and a guard rejects any number the simulator did not produce. Without internet, a template answers instead.

HONEST LIMITS

From flights alone, single nationality-months can miss by much more than the average; small markets carry the widest ranges, and the app shows them.

Three markets (China, Russia, South Korea) show more hotel guests than direct passengers, because guests also arrive on other routes. The app flags them.

A market with no route today borrows the capture of comparable markets until real data exists.

Guests arriving by road or via Dubai are held at their recent level; the flight chain cannot see them.

Score it live, on the months still to come

1. A blind test on fresh months

DCT holds back the Mar to Jul 2026 hotel actuals. We submit forecasts first, and DCT scores them with the same WMAPE. No re-fitting on the answer.

2. A real shock

Replay the Feb 2026 Gulf airspace closure through the levers and compare with what hotels saw.

3. Planner sessions

Two DCT planners bring real route questions. We time how long an answer takes and ask whether they trust it.

4. At the booth

Judges drive it: add a route, cut a market, or ask in plain English, and the chain, map and lattice update at once.



Ask: "What if 5 points fewer passengers transfer onward?" sets the lever, runs the simulator and answers with its numbers only.

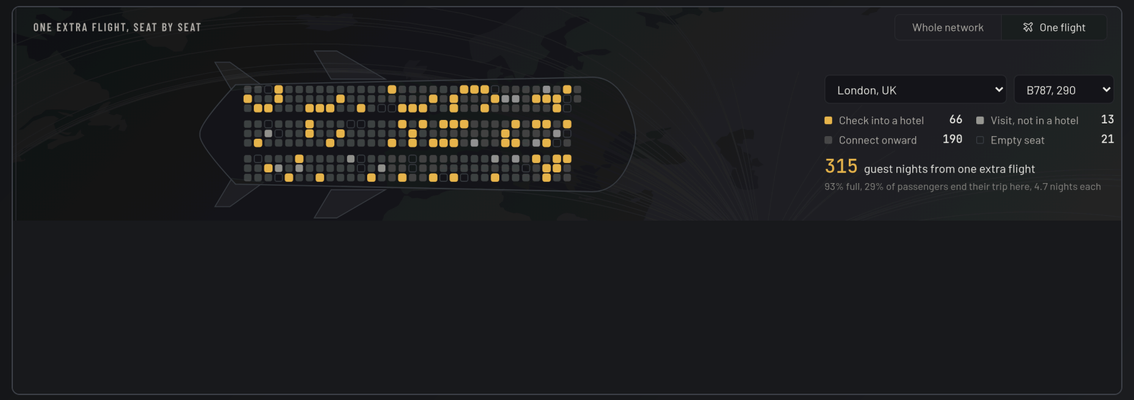
See it in the product



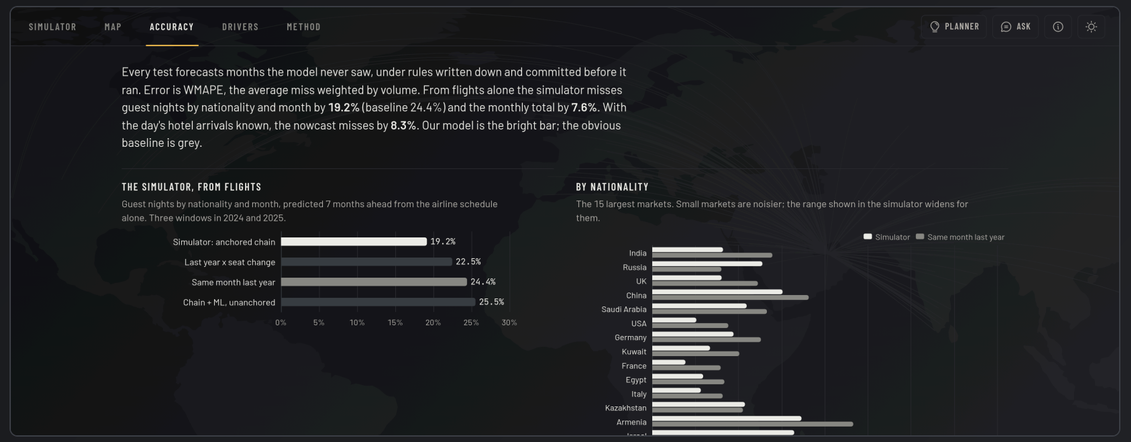
Simulator. The chain on top, every nationality by month below, the answer on the right.



Map. Every real departure city flying to AUH, month by month.



One flight. A London B787, seat by seat: who connects onward, who visits, who checks into a hotel.



Accuracy. Every test result, ours against the baseline, inside the product.